

1. pyro5-ns e servidor

5	4.513723	::1	::1	TCP	76	50123 → 9090	[SYN]	Seq=0 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM
6	4.514004	::1	::1	TCP	76	9090 → 50123	[SYN, ACK]	Seq=0 Ack=1 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM
7	4.514102	::1	::1	TCP	64	50123 → 9090	[ACK]	Seq=1 Ack=1 Win=2618880 Len=0
8	4.514531	::1	::1	TCP	178	50123 → 9090	[PSH, ACK]	Seq=1 Ack=1 Win=2618880 Len=114
9	4.514579	::1	::1	TCP	64	9090 → 50123	[ACK]	Seq=1 Ack=115 Win=2618880 Len=0
10	4.515940	::1	::1	TCP	272	9090 → 50123	[PSH, ACK]	Seq=1 Ack=115 Win=2618880 Len=208
11	4.516010	::1	::1	TCP	64	50123 → 9090	[ACK]	Seq=115 Ack=209 Win=2618624 Len=0
12	4.517322	::1	::1	TCP	291	50123 → 9090	[PSH, ACK]	Seq=115 Ack=209 Win=2618624 Len=227
13	4.517399	::1	::1	TCP	64	9090 → 50123	[ACK]	Seq=209 Ack=342 Win=2618624 Len=0
14	4.518394	::1	::1	TCP	134	9090 → 50123	[PSH, ACK]	Seq=209 Ack=342 Win=2618624 Len=70
15	4.518461	::1	::1	TCP	64	50123 → 9090	[ACK]	Seq=342 Ack=279 Win=2618624 Len=0

```
PYRO.....J.....M.# serpent utf-8 python3.2
{'handshake':'hello','object':'Pyro.NameServer'}
PYRO.....@.....*6.D..R.{...M.# serpent utf-8 python3.2
{'handshake':'hello','meta':{'methods':{'ping','lookup','list','count','remove','yplookup','set_metadata','register'},'oneway':0,'attrs':{}}}
PYRO.....M.# serpent utf-8 python3.2
('Pyro.NameServer','register','(lucas.gabriel',{'__class__':'Pyro5.core.URI','state':('PYRO','obj_23eed2eb40eb4b89b54d27274c9b8d8c',None,'localhost',50122)}),{})
PYRO.....@.....O9.Z..H..@.f|*..M.# serpent utf-8 python3.2
None
```

O servidor Pyro5 foi iniciado e estabeleceu conexão TCP com o nameserver na porta 9090

2. client

Pausa 1

5	4.527044	::1	::1	TCP	76	50784 → 9090	[SYN]	Seq=0 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM
6	4.527258	::1	::1	TCP	76	9090 → 50784	[SYN, ACK]	Seq=0 Ack=1 Win=65535 Len=0 MSS=65475 WS=256 SACK_PERM
7	4.527400	::1	::1	TCP	64	50784 → 9090	[ACK]	Seq=1 Ack=1 Win=2618880 Len=0
8	4.527693	::1	::1	TCP	178	50784 → 9090	[PSH, ACK]	Seq=1 Ack=1 Win=2618880 Len=114
9	4.527747	::1	::1	TCP	64	9090 → 50784	[ACK]	Seq=1 Ack=115 Win=2618880 Len=0

```
PYRO.....J.....M.# serpent utf-8 python3.2
{'handshake':'hello','object':'Pyro.NameServer'}
PYRO.....@.....u..I.O...
D..7..M.# serpent utf-8 python3.2
{'handshake':'hello','meta':{'methods':{'remove','lookup','list','set_metadata','register','count','ping','yplookup'},'oneway':0,'attrs':{}}}
```

Three-way handshake (aperto de mão triplo) do TCP, cliente estabelecendo conexão com o nameserver (porta 9090).

Resposta do nameserver contendo os metadados da interface, incluindo a lista completa de métodos disponíveis.

Pausa 2

10	4.528605	::1	::1	TCP	272	9090 → 50784	[PSH, ACK]	Seq=1 Ack=115 Win=2618880 Len=208
11	4.528679	::1	::1	TCP	64	50784 → 9090	[ACK]	Seq=115 Ack=209 Win=2618624 Len=0
12	4.529719	::1	::1	TCP	183	50784 → 9090	[PSH, ACK]	Seq=115 Ack=209 Win=2618624 Len=119
13	4.529789	::1	::1	TCP	64	9090 → 50784	[ACK]	Seq=209 Ack=234 Win=2618880 Len=0
14	4.530446	::1	::1	TCP	239	9090 → 50784	[PSH, ACK]	Seq=209 Ack=234 Win=2618880 Len=175
15	4.530517	::1	::1	TCP	64	50784 → 9090	[ACK]	Seq=234 Ack=384 Win=2618368 Len=0
16	4.531023	::1	::1	TCP	64	50784 → 9090	[FIN, ACK]	Seq=234 Ack=384 Win=2618368 Len=0
17	4.531089	::1	::1	TCP	64	9090 → 50784	[ACK]	Seq=384 Ack=235 Win=2618880 Len=0
18	4.531235	::1	::1	TCP	64	9090 → 50784	[FIN, ACK]	Seq=384 Ack=235 Win=2618880 Len=0
19	4.531313	::1	::1	TCP	64	50784 → 9090	[ACK]	Seq=235 Ack=385 Win=2618368 Len=0

```
PYRO.....O.....M.# serpent utf-8 python3.2
('Pyro.NameServer','lookup',('lucas.gabriel',0.0),{})
PYRO.....@.....i..J..3T..>o...M.# serpent utf-8 python3.2
{'__class__':'Pyro5.core.URI','state':('PYRO','obj_f9f2adcbaef94976b5a62b8f68e4a9cc',None,'localhost',50721)}
```

Cliente consulta o nameserver pedindo a localização do objeto 'lucas.gabriel' e o Nameserver retorna a URI do objeto.

3. Métodos

Listar

1	0.000000	::1	::1	TCP	190 50785 → 50721 [PSH, ACK] Seq=1 Ack=1 Win=10229 Len=126
2	0.000026	::1	::1	TCP	64 50721 → 50785 [ACK] Seq=1 Ack=127 Win=10229 Len=0
3	0.000428	::1	::1	TCP	132 50721 → 50785 [PSH, ACK] Seq=1 Ack=127 Win=10229 Len=68
4	0.000444	::1	::1	TCP	64 50785 → 50721 [ACK] Seq=127 Ack=69 Win=10229 Len=0

```
PYRO.....V.....M.# serpent utf-8 python3.2
('obj_f9f2adcbaef94976b5a62b8f68e4a9cc','listarTodas',(),{})
PYRO.....@.....O.B,...[D.s...M.# serpent utf-8 python3.2
[]
```

Criar

1	0.000000	::1	::1	TCP	342 50785 → 50721 [PSH, ACK] Seq=1 Ack=1 Win=10229 Len=278
2	0.000029	::1	::1	TCP	64 50721 → 50785 [ACK] Seq=1 Ack=279 Win=10228 Len=0
3	0.003601	::1	::1	TCP	134 50721 → 50785 [PSH, ACK] Seq=1 Ack=279 Win=10228 Len=70
4	0.003627	::1	::1	TCP	64 50785 → 50721 [ACK] Seq=279 Ack=71 Win=10228 Len=0

```
PYRO.....M.# serpent utf-8 python3.2
('obj_f9f2adcbaef94976b5a62b8f68e4a9cc','adicionar',({'__class__':'disciplina.Disciplina','codigo':'A21','nomeDisciplina':'a21','nomeProfessor':'a','quantidadeAlunos':21,'cargaHoraria':21,'mediaTurma':21.0}),,){})
PYRO.....@.....r.
F..e$.`&....M.# serpent utf-8 python3.2
True
```

O pyro realizando as chamadas remotas por tcp, sendo as mensagens sendo serializadas no formato serpent, chamando os métodos registrados previamente no servidor. A primeira chamada retorna uma lista vazia ([]), indicando ausência de registros, enquanto a segunda retorna True, confirmando o sucesso na inserção de um novo objeto da classe Disciplina.